

TRAINED AGENTIC CONTEXT MANAGEMENT

Anonymous authors

Paper under double-blind review

ABSTRACT

We study long context language models. Instead of training long context natively, or designing a long context harness, we train a model over the simplest possible harness: a tool to call itself with any specified prompt and a tool to read tokens in a range from the input context. We finetune Qwen3.6-35B-A3B over this harness. With only 8,000 tokens of context, our small model outperforms GPT-5.4 with 1M tokens of context on the OOLONG-synth benchmark when document length exceeds 40K tokens.

1 INTRODUCTION

As language models boast impressive breakthroughs (Luong & Lockhart, 2025; Wei, 2025; OpenAI, 2025b; Lin & Cheng, 2025; OpenAI, 2026g;f;e), increasing focus is placed on their limitations. The most salient resource constraint of a language model is its context length. Models can process and produce tokens within this limit, but the limit is hard. Model producers have identified this constraint and placed great emphasis on extending it as large as possible, with some model builders touting 10M (Dangel, 2026; Meta AI, 2025; Zhu et al., 2026) and even 100M (Magic, 2024) token context lengths (for reference, the length of the first Harry Potter book is about 100,000 tokens, the full U.S. tax code roughly 3M, and the Linux kernel source code about 300M).

While native context scaling is appealing as a concrete engineering problem, it is not the only approach to producing a model capable of processing long documents. In 2021, OpenAI fine-tuned GPT-3 to summarize sections of books, and then summarize those summaries, achieving competitive performance with human book summaries on a model with a context length of just 2,048 tokens (Wu et al., 2021). As the field has matured, agentic harnesses have become viable. MemGPT treats context similarly to an operating system, relying on LLM-invoked function calls to push relevant information in and out of the context window to external memory, effectively achieving expanded virtual context (Packer et al., 2023). More recently, Recursive Language Models introduced a harness that utilizes LLM’s coding prior to call itself in a REPL environment, achieving stronger performance than base models on retrieval and aggregation tasks both within context limits, but also beyond it, expanding effective context by $\sim 4\times$ (Zhang et al., 2025).

While handcrafted prompts and programmatic environments can be effective in solving long context problems without native long context training, it begs an obvious question: if models can, for example, solve all the problems at the International Collegiate Programming Contest¹ World Finals (OpenAI, 2025b), or more generally, implement requested features in codebases far larger than context limits (Marten et al., 2026; OpenAI, 2026d; Anthropic, 2026b), can they not be *trained* to manage context effectively, *without human-authored harness crutches*? Such model training internalizes decisions external harnesses hardcode, offering the potential to scale much more generally.

We answer this question in this paper, affirmatively. We consider the simplest possible agentic context harness, defined below. The purpose of the harness is to *extend capability*, without *restricting choices* or *instilling a prior*.

1.1 HARNESS

Our harness has two tools:

¹<https://icpc.global/>

1. `read_chunk(start, end)`: Read a slice of the document and return the decoded text of tokens `[start, end)`.
2. `spawn_subagent(subtask)`: Spawn a fresh-context copy of yourself to solve `subtask`; returns the child's `\boxed{}` answer.

Unmodified long-context prompts are provided to the model over a document accessed via the above harness. The system prompt is propagated to child agents and includes the above harness definition, a generic instruction, and potentially task-specific background, in accordance with the specified `eval2`. Subagents' user prompt is specified by the calling model.

1.2 RESULTS

We finetune Qwen3.6-35B-A3B on the above harness on a diverse synthetic data mixture and achieve the following results:

1. blah
2. blah blah

1.3 PAPER ORGANIZATION

In the next section, we survey related work to frame the contribution of the following sections. In Section 3, we describe our training methodology. In Section 4, we discuss all our experiments, and we conclude in Section 5.

2 RELATED WORK

Long-context LLM work can be broken into a few distinct categories: *native long context models*, *context compaction*, and *long context harnesses*, both trained and untrained.

2.1 NATIVE LONG CONTEXT MODELS

GPT-2 shipped with a 1,024 token context limit (Radford et al., 2019). This limit doubled in every generation through GPT-4 (Brown et al., 2020; OpenAI, 2022; 2023b), until jumping to the 100K range on GPT-4 Turbo (OpenAI, 2023a), with contemporaneous models Claude 2 (Anthropic, 2023) and later Llama 3.1 (Grattafiori et al., 2024) at similar limits. The core difficulty in scaling context length n in the classic Transformer architecture (Vaswani et al., 2017) is $\Omega(n^2)$ attention complexity, which becomes prohibitively expensive as $n \gg 100K$. State-space architectures like the mamba lineage (Gu & Dao, 2024; Dao & Gu, 2024; Lahoti et al., 2026), neural methods like Titans (Behrouz et al., 2024), and Linear Transformer (Katharopoulos et al., 2020) derivatives DeltaNet (Schlag et al., 2021) and Gated DeltaNet (Yang et al., 2025) abandon quadratic attention for recurrence. Modern architectures are hybrid, combining linear or sparse attention layers with dense attention, allowing context lengths to scale to 1M+ tokens while retaining long-context performance (Qwen Team, 2026; Kimi Team, 2026b; DeepSeek-AI, 2025; GLM-5 Team, 2026; NVIDIA, 2025; Gemma Team, 2024; Mistral AI, 2024).

2.2 CONTEXT COMPACTION

An alternative to native long context, and an inevitability when such native context fills, is a function f mapping current context x to a reduced context y where $|y| \ll |x|$. Three categories of such lossy f exist: *masking*, *summarization*, and *learned compaction*. The simplest compaction technique is masking: dropping tokens based on a predetermined rule. Clearing tool call results is a popular recommended choice (Anthropic, 2025a), achieving results competitive with summarization (DeepSeek-AI, 2025; Lindenbauer et al., 2025). Alternative approaches evict tokens based on attention statistics (Xiao et al., 2024; Zhang et al., 2023; Li et al., 2024). Summarization is the current production harness default, implemented in Claude Code, Codex, and Gemini CLI (Anthropic,

²Precise harness environment is given in Appendix A.

2025a; OpenAI, 2026a; Google, 2026; Barbaste et al., 2026). Trained compaction is a new frontier, described as part of GPT-5.1-Codex-Max (OpenAI, 2025a) and Cursor’s composer (Cassano & Rush, 2026) training. “Still” trains a model to compress KV-Cache directly (O’Neill et al., 2026).

2.3 LONG CONTEXT HARNESSES

As long as LLMs have proven useful, an evolving line of work has focused on how to expand their capabilities without touching model internals. The first efforts were known as prompt engineering (Phoenix & Taylor, 2024; Berryman & Ziegler, 2024), which, though necessary on pre-trained base models (Brown et al., 2020), became internalized through instruction-tuning and reasoning RL (Ouyang et al., 2022; Guo et al., 2025). Function-calling has led to a new wave of such efforts, where domain-specific knowledge, tools, and multi-agent systems live on a scaffold that interacts with powerful LLMs known as a “harness” (Lopopolo, 2026; Barbaste et al., 2026).

Long-context harnesses originate with systems like MemGPT (Packer et al., 2023), RAPTOR (Sarathi et al., 2024), and Chain of Agents (Zhang et al., 2024), where the overall goal is to engineer systems that decompose documents into contexts independent LLM calls can read and aggregate. Recursive Language Models is a popular recent example that allows for single-level recursive LLM calls via a programmatic environment (Zhang et al., 2025). Follow-up work expands subagent capabilities (Lumer et al., 2026), trains shared parent/child policies (Kim & Ahmad, 2026), and incorporates self-reflection (Alizadeh et al., 2026).

Memagent (Yu et al., 2026) trains a harness that walks linearly through context, using RL and a harness prompt to determine what to carry from node-to-node. Similarly to our work, they also use a small (8K token) context while processing long (as much as 3.5M token) documents. However, they test primarily on the distribution they train, compare only against other small models, and hardcode execution path and prompts in the harness.

Frontier models now utilize multi-agent systems. Kimi K2.5 trains “agent swarm” (Kimi Team, 2026a): the model is trained to call subagents via reinforcement learning. Subagent trajectories are not trained. GPT-5.6 has also trained multi-agent behavior (OpenAI, 2026c), notably utilizing 10,000 agents to solve Navier-Stokes (OpenAI, 2026e). Anthropic, DeepMind, Meta, and xAI likely do similarly, with multi-agent systems touted in respective model and product releases (Anthropic, 2025b; 2026a; OpenAI, 2026b; xAI, 2026; 2025; Meta Superintelligence Labs, 2026; Google, 2025; Gottweis et al., 2026).

3 METHODS

We deliberately use the minimal tool definitions necessary to train agentic context management. However, our harness is sufficient to allow arbitrarily complex multi-agent decompositions. Decompositions in the literature fall into roughly three categories: *tree decompositions* (Wu et al., 2021; Sarathi et al., 2024; Zhang et al., 2022), *sequential folds* (Yu et al., 2026; Zhang et al., 2024), and *agentic delegation* (Zhang et al., 2025; Kimi Team, 2026a; Anthropic, 2025b). We utilize all three strategies.

Without any training, models generally make poor use of our harness. Base Qwen3.6-35B-A3B, GPT-5.4, and Kimi K2.6 use subagent delegation sporadically, sometimes choosing to read the entire document in one context, even if context limit is specified to be less than document length, and other times delegating the exact original question to subagents. Opus 4.6 does slightly better, showing real context pressure awareness and document range subagent delegation. Unfortunately it delegates *too* much, blowing through \$300 of API credits building massive subagent trees on a 100-question eval, only to overcount³. Instead, we compare to base models supplied the entire document in context with native context limits, where performance is stronger.

In general, models have a *weak prior* for context decomposition. Pre-training is pure language continuation, and agentic post-training does not teach decomposition strategies. Where Recursive Language Models (Zhang et al., 2025) can rely on strong coding priors instilled in both pre- and post-training, decomposition on our harness needs to be taught.

³Opus 4.6 does appear to perform somewhat decently on our harness with unlimited context budget. Unfortunately, due to the massive subagent trees, running the full comparison without restriction is cost-prohibitive.

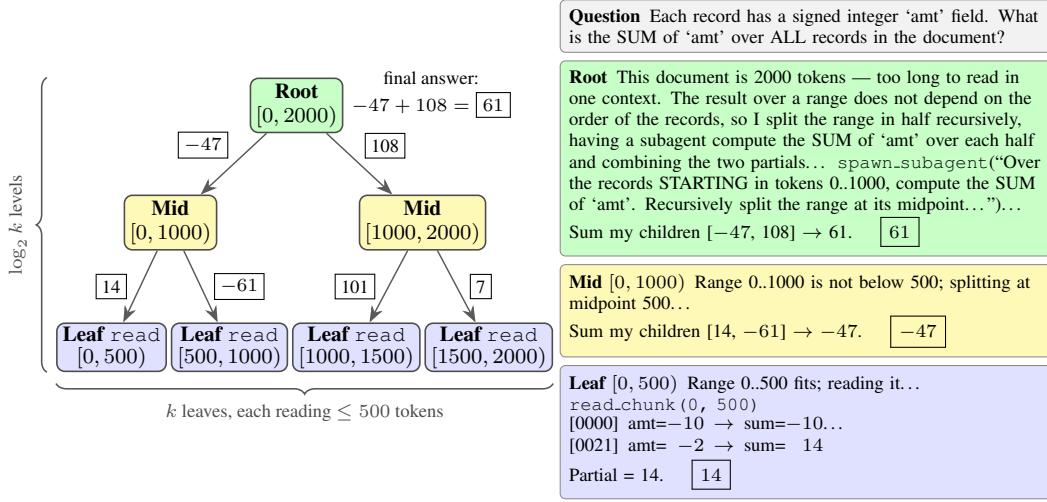


Figure 1: **Tree reduce**, for order-independent tasks (aggregation, retrieval). Non-leaf nodes delegate half their range to subagents. Leaves read one chunk and compute a leaf op which is aggregated up the tree.

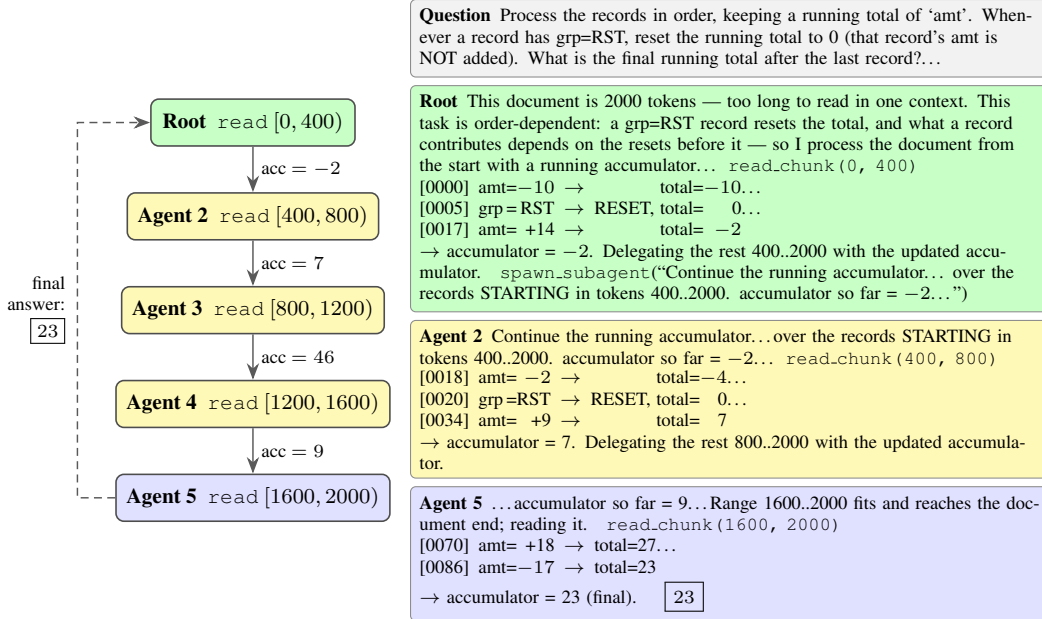


Figure 2: **Sequential fold**, for order-dependent tasks (variable tracking, running state). Each agent reads the next slice, updates the accumulator, and delegates the rest of the document to the next agent.

One idea might be to RL an open weight model on our harness. Unfortunately, exploration in LLM RL is far too weak to escape the weak decomposition prior at modest compute budgets. It is also unclear what reward to assign subagents, and if GRPO (Shao et al., 2024) or its derivatives (Yu et al., 2025; Liu et al., 2025; Zheng et al., 2025) are used, what group to compare subagent trajectories in.

We instead SFT a decomposition strategy.

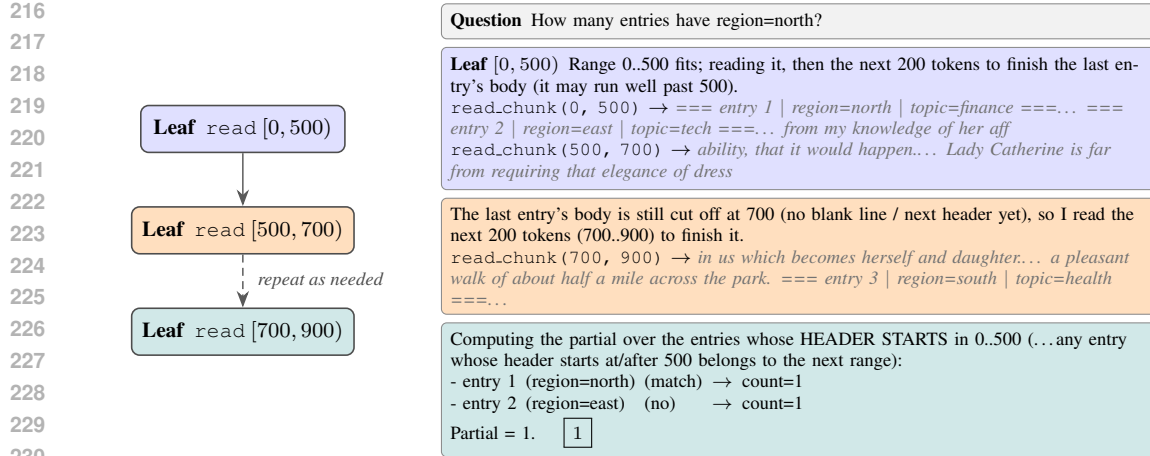


Figure 3: **Boundary handling.** A leaf owns the records whose first line starts in its range. It reads its range plus the next 200 tokens; while its last owned record is still cut off, it reads 200 more (dashed). Entry 2 starts before 500 but runs past 700, so the leaf reads to 900, where entry 3’s header appears. Entry 3 starts at or after 500, so it belongs to the next leaf and is not counted here.

4 EXPERIMENTS

5 CONCLUSION

AI USE STATEMENT

In this work, we used generative AI tools to write all our code, administer experiments, create figures, search and analyze relevant literature, format references, generate synthetic data sets, propose or refine hypotheses/brainstorming, design or provide feedback on research methodology/experiments, clean and reformat data, support qualitative and thematic data analysis, and interpret results. We have not used generative AI tools to draft parts of this research paper, and the following topics are not applicable to this work: help develop theoretical models or conceptual frameworks, formulate mathematical claims, provide critical ingredients for proving mathematical claims, assist in the writing of proofs, and assist with translation. We have reviewed our AI-assisted work, for example, by reviewing AI code and conducting manual audits of synthetic data. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REFERENCES

- Keivan Alizadeh, Parshin Shojaei, Minsik Cho, and Mehrdad Farajtabar. Recursive language models meet uncertainty: The surprising effectiveness of self-reflective program search for long context, 2026. URL <https://arxiv.org/abs/2603.15653>.
- Anthropic. Claude 2. Anthropic blog post, July 2023. URL <https://www.anthropic.com/news/claude-2>.
- Anthropic. Effective context engineering for AI agents. Engineering blog post, September 2025a. URL <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>.
- Anthropic. How we built our multi-agent research system. Engineering blog post, June 2025b. URL <https://www.anthropic.com/engineering/multi-agent-research-system>.
- Anthropic. Orchestrate teams of Claude Code sessions. Claude Code documentation, 2026a. URL <https://code.claude.com/docs/en/agent-teams>. Agent teams (experimental), introduced with Claude Opus 4.6, February 2026; accessed 2026-09-23.

- Anthropic. Introducing Claude Fable 5.1 and Claude Mythos 5.1. Anthropic blog post, September 2026b. URL <https://www.anthropic.com/claude-fable-and-mythos-5-1>.
- Paul Barbaste, Tristan Darrigol, Germain Vu, and Tom Wiltberger. Harness engineering: Anatomy, architecture, and evolution of coding agents – a source-code study of eleven systems, 2026. URL <https://arxiv.org/abs/2609.00006>.
- Ali Behrouz, Peilin Zhong, and Vahab Mirrokni. Titans: Learning to memorize at test time, 2024. URL <https://arxiv.org/abs/2501.00663>.
- John Berryman and Albert Ziegler. *Prompt Engineering for LLMs: The Art and Science of Building Large Language Model-Based Applications*. O’Reilly Media, November 2024. URL <https://www.oreilly.com/library/view/-/9781098156145>.
- Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. URL <https://arxiv.org/abs/2005.14165>.
- Federico Cassano and Sasha Rush. Training Composer for longer horizons. Cursor blog post, March 2026. URL <https://cursor.com/blog/self-summarization>.
- Justin Dangel. Introducing SubQ: The first fully subquadratic LLM. Subquadratic blog post, May 2026. URL <https://subq.ai/introducing-subq>. 12M-token research result; the shipped model is SubQ 1M-Preview.
- Tri Dao and Albert Gu. Transformers are SSMS: Generalized models and efficient algorithms through structured state space duality. In *International Conference on Machine Learning (ICML)*, 2024. URL <https://arxiv.org/abs/2405.21060>.
- DeepSeek-AI. DeepSeek-V3.2: Pushing the frontier of open large language models, 2025. URL <https://arxiv.org/abs/2512.02556>.
- Gemma Team. Gemma 2: Improving open language models at a practical size, 2024. URL <https://arxiv.org/abs/2408.00118>.
- GLM-5 Team. GLM-5: from vibe coding to agentic engineering, 2026. URL <https://arxiv.org/abs/2602.15763>.
- Google. Gemini 3 Deep Think is now available in the Gemini app. Google blog post, December 2025. URL <https://blog.google/products/gemini/gemini-3-deep-think/>.
- Google. Gemini CLI. GitHub repository, Apache-2.0, 2026. URL <https://github.com/google-gemini/gemini-cli>. Commit f50ba8608e, accessed 2026-09-23. Compression: `packages/core/src/context/chatCompressionService.ts`.
- Juraj Gottweis, Wei-Hung Weng, Alexander Daryin, et al. Accelerating scientific discovery with Co-Scientist. *Nature*, 655:487–496, 2026. URL <https://doi.org/10.1038/s41586-026-10644-y>. Preprint: arXiv:2502.18864, “Towards an AI co-scientist”.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The Llama 3 herd of models, 2024. URL <https://arxiv.org/abs/2407.21783>. Llama 3.1, 128K context.
- Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *Conference on Language Modeling (COLM)*, 2024. URL <https://arxiv.org/abs/2312.00752>.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645:633–638, 2025. URL <https://doi.org/10.1038/s41586-025-09422-z>. Preprint: arXiv:2501.12948, “DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning”.
- Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are RNNs: Fast autoregressive transformers with linear attention. In *International Conference on Machine Learning (ICML)*, 2020. URL <https://arxiv.org/abs/2006.16236>.

- Daniel Kim and Rehaan Ahmad. Reinforcing recursive language models. *alphaXiv* blog post, May 2026. URL <https://www.alphaxiv.org/blog/reinforcement-learning-for-rlms>.
- Kimi Team. Kimi K2.5: Visual agentic intelligence, 2026a. URL <https://arxiv.org/abs/2602.02276>.
- Kimi Team. Kimi K3: Open frontier intelligence. Technical report, Moonshot AI, July 2026b. URL https://github.com/MoonshotAI/Kimi-K3/blob/main/k3_tech_report.pdf. 2.8T-parameter MoE (104B active), 1M-token context; hybrid Kimi Delta Attention and full MLA (69 + 24 layers).
- Aakash Lahoti, Kevin Y. Li, Berlin Chen, Caitlin Wang, Aviv Bick, J. Zico Kolter, Tri Dao, and Albert Gu. Mamba-3: Improved sequence modeling using state space principles. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://arxiv.org/abs/2603.15569>.
- Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. SnapKV: LLM knows what you are looking for before generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL <https://arxiv.org/abs/2404.14469>.
- Hanzhao Lin and Heng-Tze Cheng. Gemini achieves gold-medal level at the International Collegiate Programming Contest World Finals. Google DeepMind blog post, September 2025. URL <https://deepmind.google/blog/gemini-achieves-gold-medal-level-at-the-international-collegiate-programming-contest>. 10 of 12 problems; would rank 2nd among university teams.
- Tobias Lindenbauer, Igor Slinko, Ludwig Felder, Egor Bogomolov, and Yaroslav Zharov. The complexity trap: Simple observation masking is as efficient as LLM summarization for agent context management. In *Fourth Deep Learning for Code Workshop (DL4C) at NeurIPS*, 2025. URL <https://arxiv.org/abs/2508.21433>.
- Zichen Liu, Changyu Chen, Wenjun Li, Penghui Qi, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. Understanding R1-Zero-like training: A critical perspective, 2025. URL <https://arxiv.org/abs/2503.20783>. Introduces Dr. GRPO.
- Ryan Lopopolo. Harness engineering: Leveraging Codex in an agent-first world. OpenAI blog post, February 2026. URL <https://openai.com/index/harness-engineering/>.
- Elias Lumer, Sahil Sen, Kevin Paul, and Vamse Kumar Subbiah. Recursive agent harnesses, 2026. URL <https://arxiv.org/abs/2606.13643>.
- Thang Luong and Edward Lockhart. Advanced version of Gemini with Deep Think officially achieves gold-medal standard at the International Mathematical Olympiad. Google DeepMind blog post, July 2025. URL <https://deepmind.google/blog/advanced-version-of-gemini-with-deep-think-officially-achieves-gold-medal-standard>. 35/42, five of six problems, graded by IMO coordinators.
- Magic. 100M token context windows. Blog post, August 2024. URL <https://magic.dev/blog/100m-token-context-windows>. LTM-2-mini; introduces the HashHop evaluation.
- Ryan Marten, Alex Shaw, Ivan Bercovich, Benedikt Droste, Tommaso Cerruti, Steven Dillmann, Mike Merrill, Andy Konwinski, and Ludwig Schmidt. Terminal-Bench: A benchmark for AI agents in terminal environments. Harbor Framework and Laude Institute, August 2026. URL <https://www.tbench.ai/>. Version 4.0.0, released 2026-08-26. Zenodo DOI 10.5281/zenodo.22105680.
- Meta AI. The Llama 4 herd: The beginning of a new era of natively multi-modal AI innovation. Blog post, April 2025. URL <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>. Llama 4 Scout, 10M-token context; model card at <https://huggingface.co/meta-llama/Llama-4-Scout-17B-16E-Instruct>.

- Meta Superintelligence Labs. Introducing Muse Spark: Scaling towards personal superintelligence. Meta AI blog post, April 2026. URL <https://ai.meta.com/blog/introducing-muse-spark-msl/>. Contemplating mode orchestrates multiple agents that reason in parallel.
- Mistral AI. Un Ministral, des Ministraux. Mistral AI blog post, October 2024. URL <https://mistral.ai/news/ministraux>. Introduces Ministral 3B and Ministral 8B; the latter uses interleaved sliding-window attention.
- NVIDIA. NVIDIA Nemotron 3: Efficient and open intelligence, 2025. URL <https://arxiv.org/abs/2512.20856>.
- Charles O’Neill, Alex Sandomirsky, Harry Partridge, Mudith Jayasekara, and Max Kirkby. Still: Amortized KV cache compaction in a single forward pass, 2026. URL <https://arxiv.org/abs/2606.07878>.
- OpenAI. Introducing ChatGPT. OpenAI blog post, November 2022. URL <https://openai.com/index/chatgpt/>. Launch of the GPT-3.5 series.
- OpenAI. New models and developer products announced at DevDay. OpenAI blog post, November 2023a. URL <https://openai.com/index/new-models-and-developer-products-announced-at-devday/>. Introduces GPT-4 Turbo with a 128K-token context window.
- OpenAI. GPT-4 technical report, 2023b. URL <https://arxiv.org/abs/2303.08774>.
- OpenAI. Building more with GPT-5.1-Codex-Max. OpenAI blog post, November 2025a. URL <https://openai.com/index/gpt-5-1-codex-max/>.
- OpenAI. Our general-purpose reasoning models solved all 12 problems at the 2025 ICPC world finals. OpenAI announcement (X), September 17, 2025, September 2025b. URL <https://x.com/OpenAI/status/1968368133024231902>. 12/12 under the five-hour contest rules; the best human team solved 11.
- OpenAI. Codex CLI. GitHub repository, Apache-2.0, 2026a. URL <https://github.com/openai/codex>. Commit 5f371ba30a, accessed 2026-09-23. Compaction prompt: `codex-rs/prompts/templates/compact/prompt.md`.
- OpenAI. Subagents. Codex documentation, 2026b. URL <https://developers.openai.com/codex/subagents>. Generally available March 2026; accessed 2026-09-23.
- OpenAI. GPT-5.6: Frontier intelligence that scales with your ambition. OpenAI blog post, July 2026c. URL <https://openai.com/index/gpt-5-6/>. Public release July 9, 2026 (Luna, Terra, Sol); introduces multi-agent mode with concurrent subagents.
- OpenAI. Introducing GPT-6 Astra. OpenAI blog post, September 2026d. URL <https://openai.com/index/gpt-6-astra/>.
- OpenAI. Finite time blowup for Navier–Stokes. Manuscript, 166 pages, with Lean formalization; announced September 8, 2026, September 2026e. URL <https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf>. Forced finite-time blow-up with bounded kinetic energy (alternative (C) of the Clay problem statement); announcement at <https://openai.com/index/navier-stokes-solution/>.
- OpenAI. Ten advances in mathematics and theoretical computer science. OpenAI blog post, August 1, 2026, August 2026f. URL <https://openai.com/index/ten-advances-in-mathematics/>. Astra; ten problems open for ten or more years, each with a Lean 4 proof certificate.
- OpenAI. An OpenAI model has disproved a central conjecture in discrete geometry. OpenAI blog post, May 2026g. URL <https://openai.com/index/model-disproves-discrete-geometry-conjecture/>. Disproof of the Erdős unit-distance conjecture: an infinite family of constructions giving a polynomial improvement over the square-grid bound, verified by external mathematicians.

- Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. URL <https://arxiv.org/abs/2203.02155>.
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems, 2023. URL <https://arxiv.org/abs/2310.08560>.
- James Phoenix and Mike Taylor. *Prompt Engineering for Generative AI: Future-Proof Inputs for Reliable AI Outputs*. O’Reilly Media, May 2024. URL <https://www.oreilly.com/library/view/prompt-engineering-for/9781098153427/>.
- Qwen Team. Qwen3.5: Towards native multimodal agents. Qwen blog post, February 2026. URL <https://qwen.ai/blog?id=qwen3.5>. First open-weight release Qwen3.5-397B-A17B; hybrid Gated DeltaNet and gated full attention (3:1) with sparse MoE.
- Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. Technical report, OpenAI, 2019. URL https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf.
- Parth Sarthi, Salman Abdullah, Aditi Tuli, Shubh Khanna, Anna Goldie, and Christopher D. Manning. RAPTOR: Recursive abstractive processing for tree-organized retrieval. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2401.18059>.
- Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *International Conference on Machine Learning (ICML)*, 2021. URL <https://arxiv.org/abs/2102.11174>.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models, 2024. URL <https://arxiv.org/abs/2402.03300>. Introduces Group Relative Policy Optimization (GRPO).
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. URL <https://arxiv.org/abs/1706.03762>.
- Alexander Wei. An experimental reasoning LLM achieves gold medal-level performance on the International Math Olympiad. OpenAI announcement (X thread), July 19, 2025, July 2025. URL https://x.com/alexwei_/status/1946477742855532918. 35/42 under contest conditions; proofs graded by three former IMO medalists.
- Jeff Wu, Long Ouyang, Daniel M. Ziegler, Nisan Stiennon, Ryan Lowe, Jan Leike, and Paul Christiano. Recursively summarizing books with human feedback, 2021. URL <https://arxiv.org/abs/2109.10862>.
- xAI. Grok 4. xAI announcement, July 2025. URL <https://x.ai/news/grok-4>. Introduces Grok 4 Heavy with parallel test-time compute.
- xAI. Multi agent. Grok API documentation, 2026. URL <https://docs.x.ai/developers/model-capabilities/text/multi-agent>. grok-4.20-multi-agent: 4 or 16 agents with a leader agent that synthesizes; released February 2026; accessed 2026-09-23.
- Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2309.17453>.

- Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving Mamba2 with delta rule. In *International Conference on Learning Representations (ICLR)*, 2025. URL <https://arxiv.org/abs/2412.06464>.
- Hongli Yu, Tinghong Chen, Jiangtao Feng, Jiangjie Chen, Weinan Dai, Qiying Yu, Ya-Qin Zhang, Wei-Ying Ma, Jingjing Liu, Mingxuan Wang, and Hao Zhou. MemAgent: Reshaping long-context LLM with multi-conv RL-based memory agent. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://arxiv.org/abs/2507.02259>.
- Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, Xin Liu, Haibin Lin, et al. DAPO: An open-source LLM reinforcement learning system at scale. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL <https://arxiv.org/abs/2503.14476>.
- Alex L. Zhang, Tim Kraska, and Omar Khattab. Recursive language models, 2025. URL <https://arxiv.org/abs/2512.24601>. v1 December 2025; revised May 2026 with the SFT-trained RLM-Qwen3-8B.
- Yusen Zhang, Ansong Ni, Ziming Mao, Chen Henry Wu, Chenguang Zhu, Budhaditya Deb, Ahmed H. Awadallah, Dragomir Radev, and Rui Zhang. Summ^N: A multi-stage summarization framework for long input dialogues and documents. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022. URL <https://arxiv.org/abs/2110.10150>.
- Yusen Zhang, Ruoxi Sun, Yanfei Chen, Tomas Pfister, Rui Zhang, and Sercan Ö. Arık. Chain of agents: Large language models collaborating on long-context tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL <https://arxiv.org/abs/2406.02818>.
- Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. H₂O: Heavy-hitter oracle for efficient generative inference of large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL <https://arxiv.org/abs/2306.14048>.
- Chujie Zheng, Shixuan Liu, Mingze Li, Xiong-Hui Chen, Bowen Yu, Chang Gao, Kai Dang, Yuqiong Liu, Rui Men, An Yang, Jingren Zhou, and Junyang Lin. Group sequence policy optimization, 2025. URL <https://arxiv.org/abs/2507.18071>.
- Zheqing Zhu, Christopher Wu, Yi Wan, Yiqing Cai, Qiwei He, Ruihao Zhu, Chang Zen Tee, Lexin Chen, Pu Thavikulwat, and Jared Ee. Pokee-Isaac 28B: A 10M-token context efficient agentic model. Technical report, Pokee AI, August 2026. URL <https://console.pokee.ai/pokee-isaac-28b-v0-technical-report.pdf>.

A HARNESS AND PROMPTS

You may include other additional sections here.